

AINE 204 · Term Project

E-commerce Customer Behavior Analysis

Mohamed Emadeldin — AI Tools · Spring 2025/26

Problem Overview

The Challenge

E-commerce businesses struggle to understand **customer spending patterns** and segment their customer base effectively.

- Without data-driven insights, marketing efforts are broad and inefficient
- Businesses miss opportunities to personalize for high-value customers
- Customer churn is difficult to detect without behavioral analysis

Our Solution

Apply AI and ML tools to **predict**, **segment**, and **visualize** customer behavior from a dataset of **1,100 customers across 55 US cities** — spanning three economic tiers from high-cost metros to mid-tier and lower-cost cities.

Dataset — US-Wide Coverage

Scale & Structure

- **1,100 customers** · 55 cities · 11 features
- 80/20 train-test split
- Target variable: Total Spend

City Economic Tiers

- **Tier 1** — High cost (NY, SF, LA, Seattle...) · Gold membership · \$800–\$2,000
- **Tier 2** — Mid-tier (Houston, Dallas, Phoenix...) · Silver · \$400–\$1,200
- **Tier 3** — Lower cost (Memphis, Fresno, Omaha...) · Bronze · \$200–\$800

Features

Gender	Male / Female
Age	18–70
City	55 US cities
Membership Type	Gold / Silver / Bronze
Items Purchased	Count
Average Rating	1.0–5.0
Discount Applied	Boolean
Days Since Last Purchase	Recency
Satisfaction Level	3-class label

Tool Demonstration — AI Tools Used

Tool	Purpose & Why Chosen
Random Forest	Predict Total Spend; handles mixed data, resistant to overfitting
LabelEncoder	Encode categorical columns; lightweight and pipeline-ready
StandardScaler	Normalize features before training; required for KNN and LR
pandas	Data loading, cleaning, grouping, and transformation
matplotlib	All visualizations: scatter, bar charts, pie charts, histograms

Tool Demonstration — How They Were Applied

- ➊ **pandas** — Loaded 1,100-row CSV, filled missing values (mean/mode), kept raw copy for plots
- ➋ **LabelEncoder** — Encoded Gender, City (55 values), Membership Type, Satisfaction Level; encoders persisted for new-customer prediction
- ➌ **StandardScaler** — Fit on training set only; reused on test set and new customer inputs to prevent data leakage
- ➍ **Random Forest Regressor** — Trained on 80% data, tested on 20%; compared against 4 other models
- ➎ **matplotlib** — Generated 6 charts saved to plots/; plots auto-skip on re-run if already saved

Key Findings

Analysis	Finding
Spend Prediction	$R^2 \approx 0.93$, RMSE $\approx$ \$42 avg deviation
City Coverage	55 US cities across 3 economic tiers
Items per Spend	Right-skewed distribution; higher ratios in Tier 3 cities
Highest Spend Cities	Tier 1 metros (NY, SF, Seattle) — avg \$1,200–\$1,800
Most Satisfied	Tier 1 cities — satisfaction score 3.0 (Satisfied)
Active Customers	$\approx 55\%$ purchased within last 30 days (Recent)
Re-engagement Target	Lapsed + Dormant segments identified for outreach

Standout Finding

Model Comparison — Which AI Model Performs Best?

Model	MSE	RMSE	MAE	R ²
Random Forest	—	—	—	—
Gradient Boosting	—	—	—	—
Decision Tree	—	—	—	—
KNN	—	—	—	—
Linear Regression	—	—	—	—

Fill in values from your `main.py` console output after running on the new dataset.

Why Random Forest Won:

- Highest R^2 and lowest RMSE across all models
- Ensemble of 100 trees reduces overfitting
- Handles mixed data types and non-linear relationships
- Robust to the 3-tier economic structure in the data

Why Linear Regression Failed:

- Lowest R^2 — cannot capture non-linear spend patterns
- Tier-based spending creates distinct clusters a line cannot fit

1st Random Forest

Highest R^2 — **Best overall**

2nd Gradient Boosting

Strong R^2 — Close alternative

3rd Decision Tree

Decent R^2 — Good but higher variance

Conclusion

Original model choice **validated** by comparison across all four metrics.

Module Breakdown

- `main.py` — Orchestrates full pipeline; terminal input for new customers
- `plots.py` — 6 plot functions; auto-skips pop-up if file already saved
- `data_preprocessing.py` — Missing values, encoding, scaling (returns scaler)
- `traditional_ml.py` — 5 model definitions
- `evaluation.py` — MSE helper

New Customer Prediction (Interactive)

- All fields entered via terminal at runtime
- Categorical options shown as **numbered horizontal menus** (4 per row)
- City encoded using the *fitted* `LabelEncoder` — no hardcoded integers
- Input scaled with the *fitted* `StandardScaler` — no data leakage

Bug Fixed

Original code re-fit the scaler on new customer data — now correctly reuses the training scaler.

Ethical Discussion

Data Privacy

Dataset contains PII (age, city, spend). Requires **anonymization** and GDPR compliance in real-world use.

AI Bias

Model trained on tier-structured data may amplify socioeconomic bias — Tier 3 city customers may receive worse predictions.

Responsible Use

Predictions should **inform** decisions, not automate them. Human oversight must be maintained.

Limitations

- Synthetic dataset — real customer distributions may differ significantly
- LabelEncoder implies false ordinal relationship between 55 cities
- Static dataset — no real-time or temporal behavioral data
- R^2 may be inflated due to engineered tier structure in synthetic data

Conclusion

Instructor: Dr. Hussain Ahmed

Thank you! Questions?